

```
In [1]: !pip install fg-data-profiling pandas numpy --quiet
```

```
In [92]: import pandas as pd
import numpy as np
import data_profiling
from data_profiling import ProfileReport
```

```
In [127... #Loading data set to be reviewed and cleaned
df = pd.read_csv('My_Projects/Urban_Forestry/Cityworks_WO/Cityworks_WOs_2022.csv')
```

```
In [128... #Generating an initial Profile Report of the data set to obtain a baseline understanding of the structur
profile = ProfileReport(df, title="Data Profile Report", explorative=True)
profile.to_notebook_iframe()
```

Summarize dataset: 0%| | 0/5 [00:00<?, ?it/s]

|      |                               |
|------|-------------------------------|
| 0%   | 0/26 [00:00<?, ?it/s]         |
| 8%   | 2/26 [00:00<00:01, 15.32it/s] |
| 15%  | 4/26 [00:00<00:01, 17.75it/s] |
| 23%  | 6/26 [00:02<00:12, 1.61it/s]  |
| 35%  | 9/26 [00:04<00:10, 1.62it/s]  |
| 50%  | 13/26 [00:05<00:04, 2.78it/s] |
| 58%  | 15/26 [00:06<00:04, 2.29it/s] |
| 100% | 26/26 [00:06<00:00, 3.99it/s] |

Generate report structure: 0%| | 0/1 [00:00<?, ?it/s]

Render HTML: 0%| | 0/1 [00:00<?, ?it/s]

Data Profile Report



Overview

Overview

Alerts 31

Reproduction

Dataset statistics

|                               |         |
|-------------------------------|---------|
| Number of variables           | 26      |
| Number of observations        | 2958    |
| Missing cells                 | 3918    |
| Missing cells (%)             | 5.1%    |
| Duplicate rows                | 0       |
| Duplicate rows (%)            | 0.0%    |
| Total size in memory          | 4.1 MiB |
| Average record size in memory | 1.4 KiB |

Variable types

|             |    |
|-------------|----|
| Numeric     | 4  |
| Categorical | 12 |
| DateTime    | 5  |
| Text        | 5  |

Variables

Select Columns

## WorkOrderId

```
In [129... #Printing another structural overview of the data set  
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2958 entries, 0 to 2957
Data columns (total 26 columns):
#   Column                Non-Null Count  Dtype
---  -
0   WorkOrderId           2958 non-null   int64
1   Description            2958 non-null   object
2   Priority                2958 non-null   int64
3   Status                 2958 non-null   object
4   Submit To              2958 non-null   object
5   Proj Start Date        2958 non-null   object
6   Proj Finish Date       2958 non-null   object
7   Actual Start           1422 non-null   object
8   Actual Finish          2956 non-null   object
9   Shop                   2873 non-null   object
10  Map Page               2958 non-null   object
11  Address                 2958 non-null   object
12  Location                2958 non-null   object
13  Category                2958 non-null   object
14  Date Initiated          2958 non-null   object
15  Project                 1361 non-null   object
16  Total Cost              2958 non-null   float64
17  Labor Cost              2958 non-null   float64
18  Comments                2797 non-null   object
19  Zone                    2955 non-null   object
20  Units Accomplished      2958 non-null   int64
21  Request Ids             2826 non-null   object
22  Resolution               2952 non-null   object
23  SR Problem Code         2826 non-null   object
24  SR Description          2826 non-null   object
25  SR Priority              2826 non-null   float64
dtypes: float64(3), int64(3), object(20)
memory usage: 601.0+ KB

```

```

In [130... #Calculating the sum of all missing values for each column
df.isna().sum()

```

```
Out[130... WorkOrderId          0
           Description         0
           Priority             0
           Status              0
           Submit To           0
           Proj Start Date     0
           Proj Finish Date    0
           Actual Start       1536
           Actual Finish       2
           Shop                85
           Map Page            0
           Address             0
           Location            0
           Category            0
           Date Initiated      0
           Project             1597
           Total Cost          0
           Labor Cost          0
           Comments            161
           Zone                3
           Units Accomplished  0
           Request Ids         132
           Resolution          6
           SR Problem Code     132
           SR Description      132
           SR Priority          132
           dtype: int64
```

```
In [131... #Reviewing the first 10 rows of the data set
df.head(10)
```

Out[131...

|   | WorkOrderId | Description                                     | Priority | Status | Submit To         | Proj<br>Start<br>Date         | Proj<br>Finish<br>Date        | Actual<br>Start               | Actual<br>Finish              | Shop | ... | Total<br>Cost | Labor<br>Cost | Comments                                                       |
|---|-------------|-------------------------------------------------|----------|--------|-------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|------|-----|---------------|---------------|----------------------------------------------------------------|
| 0 | 362924      | TRIM BY<br>CONTRACTOR                           | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-<br>01-04<br>07:24<br>AM | 2022-<br>02-03<br>07:24<br>AM | 2022-<br>04-01<br>12:00<br>PM | 2022-<br>05-11<br>12:00<br>PM | C    | ... | 147.0         | 147.0         | Invoice<br>#128168.                                            |
| 1 | 362925      | EMERGENCY<br>DEBRIS PICK-<br>UP                 | 1        | CLOSED | UF, IN-HOUSE      | 2022-<br>01-04<br>07:45<br>AM | 2022-<br>01-05<br>07:45<br>AM | NaN                           | 2022-<br>01-11<br>10:39<br>AM | K    | ... | 0.0           | 0.0           | From<br>Request<br>185199:\n<br>Code: Tree<br>Removal\n<br>... |
| 2 | 362926      | REMOVAL &<br>STUMP<br>GRINDING BY<br>CONTRACTOR | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-<br>01-04<br>07:45<br>AM | 2022-<br>02-03<br>07:45<br>AM | 2022-<br>04-01<br>12:00<br>PM | 2022-<br>08-17<br>12:00<br>PM | F    | ... | 283.0         | 283.0         | Invoice<br>#131309.                                            |
| 3 | 362927      | STUMP<br>GRINDING BY<br>CONTRACTOR              | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-<br>01-04<br>07:55<br>AM | 2023-<br>01-04<br>07:55<br>AM | 2022-<br>04-01<br>12:00<br>PM | 2022-<br>08-17<br>12:00<br>PM | F    | ... | 52.0          | 52.0          | Invoice<br>#131309.                                            |
| 4 | 362928      | REMOVAL BY<br>CONTRACTOR                        | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-<br>01-04<br>08:00<br>AM | 2022-<br>02-03<br>08:00<br>AM | 2022-<br>04-01<br>12:00<br>PM | 2023-<br>01-31<br>12:00<br>PM | B    | ... | 246.0         | 246.0         | INV<br>135853                                                  |
| 5 | 362929      | STUMP<br>GRINDING BY<br>CONTRACTOR              | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-<br>01-04<br>08:11<br>AM | 2023-<br>01-04<br>08:11<br>AM | 2022-<br>04-01<br>12:00<br>PM | 2022-<br>08-17<br>12:00<br>PM | F    | ... | 89.0          | 89.0          | Invoice<br>#131309.                                            |
| 6 | 362930      | STUMP<br>GRINDING BY<br>CONTRACTOR              | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-<br>01-04<br>08:25<br>AM | 2023-<br>01-04<br>08:25<br>AM | 2022-<br>04-01<br>12:00<br>PM | 2022-<br>08-17<br>12:00<br>PM | F    | ... | 183.0         | 183.0         | Invoice<br>#131309.                                            |
| 7 | 362931      | STUMP<br>GRINDING BY<br>CONTRACTOR              | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-<br>01-04<br>08:31<br>AM | 2023-<br>01-04<br>08:31<br>AM | 2022-<br>04-01<br>12:00<br>PM | 2022-<br>08-17<br>12:00<br>PM | F    | ... | 115.0         | 115.0         | Invoice<br>#131309.                                            |

|   | WorkOrderId | Description                   | Priority | Status | Submit To    | Proj<br>Start<br>Date | Proj<br>Finish<br>Date | Actual<br>Start | Actual<br>Finish    | Shop | ... | Total<br>Cost | Labor<br>Cost | Comments                                        |
|---|-------------|-------------------------------|----------|--------|--------------|-----------------------|------------------------|-----------------|---------------------|------|-----|---------------|---------------|-------------------------------------------------|
| 8 | 362932      | DEBRIS PICK-UP                | 2        | CLOSED | UF, IN-HOUSE | 2022-01-04 08:54 AM   | 2022-01-14 08:54 AM    | NaN             | 2022-01-20 10:27 AM | K    | ... | 0.0           | 0.0           | From Request 185191:\n Code: Tree Removal\n ... |
| 9 | 362933      | EMERGENCY REMOVAL BY IN-HOUSE | 1        | CLOSED | UF, IN-HOUSE | 2022-01-04 08:55 AM   | 2022-01-05 08:55 AM    | NaN             | 2022-01-07 02:05 PM | A    | ... | 336.0         | 336.0         | From Request 185196:\n Code: Tree Removal\n ... |

10 rows × 26 columns

```
In [132... #Standardizing column headings to remove spaces and make all lower case
df.columns = (df.columns.str.replace(' ', '_').str.lower()).str.strip()
df
```

Out[132...

|             | workorderid | description                                     | priority | status | submit_to         | proj_start_date        | proj_finish_date       | actual_start           | actual_finish          | shc |
|-------------|-------------|-------------------------------------------------|----------|--------|-------------------|------------------------|------------------------|------------------------|------------------------|-----|
| <b>0</b>    | 362924      | TRIM BY<br>CONTRACTOR                           | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>07:24 AM | 2022-02-03<br>07:24 AM | 2022-04-01<br>12:00 PM | 2022-05-11<br>12:00 PM |     |
| <b>1</b>    | 362925      | EMERGENCY<br>DEBRIS PICK-UP                     | 1        | CLOSED | UF, IN-HOUSE      | 2022-01-04<br>07:45 AM | 2022-01-05<br>07:45 AM | NaN                    | 2022-01-11<br>10:39 AM |     |
| <b>2</b>    | 362926      | REMOVAL &<br>STUMP<br>GRINDING BY<br>CONTRACTOR | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>07:45 AM | 2022-02-03<br>07:45 AM | 2022-04-01<br>12:00 PM | 2022-08-17<br>12:00 PM |     |
| <b>3</b>    | 362927      | STUMP<br>GRINDING BY<br>CONTRACTOR              | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>07:55 AM | 2023-01-04<br>07:55 AM | 2022-04-01<br>12:00 PM | 2022-08-17<br>12:00 PM |     |
| <b>4</b>    | 362928      | REMOVAL BY<br>CONTRACTOR                        | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>08:00 AM | 2022-02-03<br>08:00 AM | 2022-04-01<br>12:00 PM | 2023-01-31<br>12:00 PM |     |
| ...         | ...         | ...                                             | ...      | ...    | ...               | ...                    | ...                    | ...                    | ...                    | ... |
| <b>2953</b> | 365918      | TREE/SEEDLING<br>PLANTING                       | 4        | CLOSED | UF,<br>CONTRACTOR | 2022-12-29<br>07:14 AM | 2023-12-29<br>07:14 AM | NaN                    | 2022-12-29<br>12:00 PM |     |
| <b>2954</b> | 365919      | TRIM BY<br>CONTRACTOR                           | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-12-29<br>07:31 AM | 2023-01-28<br>07:31 AM | 2023-03-13<br>12:00 PM | 2025-03-21<br>09:08 AM |     |
| <b>2955</b> | 365920      | TREE/SEEDLING<br>PLANTING                       | 4        | CLOSED | UF,<br>CONTRACTOR | 2022-12-30<br>08:15 AM | 2023-12-30<br>08:15 AM | NaN                    | 2022-12-30<br>12:00 PM |     |

|      | workorderid | description        | priority | status | submit_to      | proj_start_date     | proj_finish_date    | actual_start        | actual_finish       | shc |
|------|-------------|--------------------|----------|--------|----------------|---------------------|---------------------|---------------------|---------------------|-----|
| 2956 | 365921      | TRIM BY CONTRACTOR | 3        | CLOSED | UF, CONTRACTOR | 2022-12-30 12:45 PM | 2023-01-29 12:45 PM | 2023-03-13 12:00 PM | 2023-04-11 12:00 PM |     |
| 2957 | 365922      | TRIM BY IN-HOUSE   | 2        | CLOSED | UF, IN-HOUSE   | 2022-12-30 01:13 PM | 2023-01-09 01:13 PM | NaN                 | 2023-01-19 02:25 PM | C   |

2958 rows × 26 columns

```
In [133... #Dropping unnecessary columns - units_accomplished and labor_cost
df = df.drop(columns = ['units_accomplished', 'labor_cost'])
```

```
In [134... #Filtering out all null values from request_ids
df['request_ids'].isnull()
```

```
Out[134... 0      False
1      False
2       True
3       True
4      False
...
2953   False
2954   False
2955   False
2956   False
2957   False
Name: request_ids, Length: 2958, dtype: bool
```

```
In [135... #Displaying the filter as a dataframe to review the rows with null values in request_ids
df[df['request_ids'].isnull()]
```

Out[135...

|             | workorderid | description                                     | priority | status | submit_to         | proj_start_date        | proj_finish_date       | actual_start           | actual_finish          | shop |
|-------------|-------------|-------------------------------------------------|----------|--------|-------------------|------------------------|------------------------|------------------------|------------------------|------|
| <b>2</b>    | 362926      | REMOVAL &<br>STUMP<br>GRINDING BY<br>CONTRACTOR | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>07:45 AM | 2022-02-03<br>07:45 AM | 2022-04-01<br>12:00 PM | 2022-08-17<br>12:00 PM | F    |
| <b>3</b>    | 362927      | STUMP<br>GRINDING BY<br>CONTRACTOR              | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>07:55 AM | 2023-01-04<br>07:55 AM | 2022-04-01<br>12:00 PM | 2022-08-17<br>12:00 PM | F    |
| <b>5</b>    | 362929      | STUMP<br>GRINDING BY<br>CONTRACTOR              | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>08:11 AM | 2023-01-04<br>08:11 AM | 2022-04-01<br>12:00 PM | 2022-08-17<br>12:00 PM | F    |
| <b>6</b>    | 362930      | STUMP<br>GRINDING BY<br>CONTRACTOR              | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>08:25 AM | 2023-01-04<br>08:25 AM | 2022-04-01<br>12:00 PM | 2022-08-17<br>12:00 PM | F    |
| <b>7</b>    | 362931      | STUMP<br>GRINDING BY<br>CONTRACTOR              | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>08:31 AM | 2023-01-04<br>08:31 AM | 2022-04-01<br>12:00 PM | 2022-08-17<br>12:00 PM | F    |
| ...         | ...         | ...                                             | ...      | ...    | ...               | ...                    | ...                    | ...                    | ...                    | ...  |
| <b>2058</b> | 365005      | EMERGENCY<br>TRIM BY IN-<br>HOUSE               | 1        | CLOSED | UF, IN-HOUSE      | 2022-08-11<br>08:30 AM | 2022-08-12<br>08:30 AM | NaN                    | 2022-08-13<br>12:00 PM | 07   |
| <b>2129</b> | 365079      | REMOVAL &<br>STUMP<br>GRINDING BY<br>CONTRACTOR | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-08-17<br>02:41 PM | 2022-09-16<br>02:41 PM | NaN                    | 2023-01-11<br>07:47 PM | I    |
| <b>2574</b> | 365533      | EMERGENCY<br>REMOVAL BY<br>IN-HOUSE             | 1        | CLOSED | UF, IN-HOUSE      | 2022-10-12<br>04:43 PM | 2022-10-13<br>04:43 PM | NaN                    | 2022-10-14<br>03:00 PM | 01   |
| <b>2575</b> | 365534      | STUMP<br>GRINDING BY<br>CONTRACTOR              | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-10-12<br>04:49 PM | 2023-10-12<br>04:49 PM | 2023-03-13<br>12:00 PM | 2023-04-22<br>12:00 PM | J    |
| <b>2622</b> | 365581      | TRIM BY IN-<br>HOUSE                            | 2        | CLOSED | UF, IN-HOUSE      | 2022-10-21<br>09:08 AM | 2022-11-20<br>09:08 AM | NaN                    | 2022-12-28<br>09:35 AM | 02   |

132 rows × 24 columns

```
In [136... #Dropping all rows where request_ids is null (NaN). Data owner reports these are phantom/duplicate records  
df = df.dropna(subset=['request_ids'])
```

```
In [137... #Recalculating the sum of all missing values for each column  
df.isna().sum()
```

```
Out[137... workorderid          0  
description          0  
priority             0  
status               0  
submit_to           0  
proj_start_date      0  
proj_finish_date     0  
actual_start        1504  
actual_finish        2  
shop                 82  
map_page             0  
address              0  
location             0  
category             0  
date_initiated       0  
project             1562  
total_cost           0  
comments            137  
zone                 1  
request_ids          0  
resolution           6  
sr_problem_code      0  
sr_description       0  
sr_priority          0  
dtype: int64
```

```
In [138... #Replacing null values in actual_start with values from proj_start_date  
#df['actual_start'] = np.where(df['actual_start'].isnull(), df['proj_start_date'], df['actual_start'])  
df['actual_start'] = df['actual_start'].fillna(df['proj_start_date'])
```

```
In [139... #Recalculating the sum of all missing values for each column  
df.isna().sum()
```

```
Out[139...] workorderid      0
            description    0
            priority       0
            status        0
            submit_to     0
            proj_start_date 0
            proj_finish_date 0
            actual_start   0
            actual_finish  2
            shop          82
            map_page      0
            address       0
            location      0
            category      0
            date_initiated 0
            project       1562
            total_cost     0
            comments      137
            zone          1
            request_ids    0
            resolution     6
            sr_problem_code 0
            sr_description  0
            sr_priority    0
            dtype: int64
```

```
In [140...] #Filtering out all null values from resolution
            df['resolution'].isnull()
```

```
Out[140...] 0      False
            1      False
            4      False
            8      False
            9      False
            ...
            2953   False
            2954   False
            2955   False
            2956   False
            2957   False
            Name: resolution, Length: 2826, dtype: bool
```

```
In [141... #Displaying the filter as a dataframe to review the rows with null values in resolution
df[df['resolution'].isnull()]
```

Out[141...

|             | workorderid | description                            | priority | status | submit_to      | proj_start_date     | proj_finish_date    | actual_start        | actual_finish       | shop |
|-------------|-------------|----------------------------------------|----------|--------|----------------|---------------------|---------------------|---------------------|---------------------|------|
| <b>386</b>  | 363316      | EMERGENCY TRIM BY IN-HOUSE             | 1        | CLOSED | UF, IN-HOUSE   | 2022-02-21 07:41 AM | 2022-02-22 07:41 AM | 2022-02-21 07:41 AM | 2022-02-23 02:58 PM | G    |
| <b>1007</b> | 363943      | REMOVAL & STUMP GRINDING BY CONTRACTOR | 3        | ISSUED | UF, CONTRACTOR | 2022-05-05 04:57 PM | 2022-06-04 04:57 PM | 2022-08-05 12:00 PM | NaN                 | H    |
| <b>1419</b> | 364361      | TRIM BY IN-HOUSE                       | 2        | CLOSED | UF, IN-HOUSE   | 2022-06-06 08:16 AM | 2022-07-06 08:16 AM | 2022-06-06 08:16 AM | 2022-10-21 02:56 PM | 05   |
| <b>1534</b> | 364478      | TRIM BY IN-HOUSE                       | 2        | CLOSED | UF, IN-HOUSE   | 2022-06-13 01:27 PM | 2022-07-13 01:27 PM | 2022-06-13 01:27 PM | 2022-11-14 03:11 PM | 09   |
| <b>1923</b> | 364869      | WATER TREES                            | 4        | CLOSED | UF, IN-HOUSE   | 2022-08-09 05:29 PM | 2022-08-23 05:29 PM | 2022-08-09 05:29 PM | 2022-09-27 02:58 PM | 03   |
| <b>2457</b> | 365411      | WATER TREES                            | 4        | CLOSED | UF, IN-HOUSE   | 2022-10-11 02:58 PM | 2022-10-25 02:58 PM | 2022-10-11 02:58 PM | 2022-10-03 03:20 PM | 04   |

6 rows × 24 columns

```
In [142... #Replacing null values in resolution with known values
df['resolution'] = df['resolution'].fillna('00 - Work completed by in-house or contractor.')
```

```
In [143... #Confirming null values in resolution have been replaced with the known values
df[df['resolution'].isnull()]
```

Out[143... **workorderid** **description** **priority** **status** **submit\_to** **proj\_start\_date** **proj\_finish\_date** **actual\_start** **actual\_finish** **shop** ... **date\_initi**

---

0 rows × 24 columns

```
In [144... #Resetting the index back to a clean sequence  
df = df.reset_index(drop=True)
```

```
In [145... #Filtering out all null values from shop  
df['shop'].isnull()
```

```
Out[145... 0      False  
1      False  
2      False  
3      False  
4      False  
      ...  
2821   False  
2822   False  
2823   False  
2824   False  
2825   False  
Name: shop, Length: 2826, dtype: bool
```

```
In [146... #Displaying the filter as a dataframe to review the rows with null values in shop  
df[df['shop'].isnull()]
```

Out[146...

|             | workorderid | description                            | priority | status | submit_to      | proj_start_date     | proj_finish_date    | actual_start        | actual_finish       | shc |
|-------------|-------------|----------------------------------------|----------|--------|----------------|---------------------|---------------------|---------------------|---------------------|-----|
| <b>144</b>  | 363122      | TREE/SEEDLING PLANTING                 | 4        | CLOSED | UF, CONTRACTOR | 2022-01-26 09:17 AM | 2023-01-26 09:17 AM | 2022-01-26 09:17 AM | 2022-01-26 12:00 PM | Na  |
| <b>212</b>  | 363200      | REMOVAL & STUMP GRINDING BY CONTRACTOR | 3        | CLOSED | UF, CONTRACTOR | 2022-02-03 11:54 AM | 2022-03-05 11:54 AM | 2022-05-02 12:00 PM | 2022-10-20 12:00 PM | Na  |
| <b>354</b>  | 363378      | TREE/SEEDLING PLANTING                 | 4        | CLOSED | UF, CONTRACTOR | 2022-03-02 06:45 AM | 2023-03-02 06:45 AM | 2022-03-02 06:45 AM | 2022-03-02 12:00 PM | Na  |
| <b>486</b>  | 363511      | EMERGENCY DEBRIS PICK-UP               | 1        | CLOSED | UF, IN-HOUSE   | 2022-03-22 08:38 AM | 2022-03-23 08:38 AM | 2022-03-22 08:38 AM | 2022-03-22 10:47 AM | Na  |
| <b>487</b>  | 363512      | REMOVAL & STUMP GRINDING BY CONTRACTOR | 3        | CLOSED | UF, CONTRACTOR | 2022-03-22 08:46 AM | 2022-04-21 08:46 AM | 2022-07-01 12:00 PM | 2022-09-08 12:00 PM | Na  |
| ...         | ...         | ...                                    | ...      | ...    | ...            | ...                 | ...                 | ...                 | ...                 | ... |
| <b>2776</b> | 365873      | REMOVAL BY CONTRACTOR                  | 3        | CLOSED | UF, CONTRACTOR | 2023-01-09 12:00 PM | 2023-06-09 12:00 PM | 2023-01-09 12:00 PM | 2023-01-20 12:00 PM | Na  |
| <b>2777</b> | 365874      | REMOVAL BY CONTRACTOR                  | 3        | CLOSED | UF, CONTRACTOR | 2023-01-09 12:00 PM | 2023-06-09 12:00 PM | 2023-01-09 12:00 PM | 2023-01-20 12:00 PM | Na  |
| <b>2782</b> | 365879      | REMOVAL & STUMP GRINDING BY CONTRACTOR | 3        | CLOSED | UF, CONTRACTOR | 2023-01-09 12:00 PM | 2023-06-09 12:00 PM | 2023-01-09 12:00 PM | 2023-02-22 12:00 PM | Na  |

|             | workorderid | description                                     | priority | status | submit_to         | proj_start_date        | proj_finish_date       | actual_start           | actual_finish          | shc |
|-------------|-------------|-------------------------------------------------|----------|--------|-------------------|------------------------|------------------------|------------------------|------------------------|-----|
| <b>2783</b> | 365880      | REMOVAL &<br>STUMP<br>GRINDING BY<br>CONTRACTOR | 3        | CLOSED | UF,<br>CONTRACTOR | 2023-01-09<br>12:00 PM | 2023-06-09<br>12:00 PM | 2023-01-09<br>12:00 PM | 2023-01-20<br>12:00 PM | Na  |
| <b>2784</b> | 365881      | REMOVAL BY<br>CONTRACTOR                        | 3        | CLOSED | UF,<br>CONTRACTOR | 2023-01-09<br>12:00 PM | 2023-06-09<br>12:00 PM | 2023-01-09<br>12:00 PM | 2023-01-20<br>12:00 PM | Na  |

82 rows × 24 columns

```
In [147... #Displaying address and map_page columns to help determine shop value. Reviewing first 60 records.
df_columns = df[['address', 'map_page', 'shop', 'workorderid']]
df_columns[df_columns['shop'].isnull()].head(60)
```

Out[147...

|     | address                              | map_page | shop | workorderid |
|-----|--------------------------------------|----------|------|-------------|
| 144 | 750 BUFFALO RUN                      | 570Q     | NaN  | 363122      |
| 212 | 14334 LOURDES DR                     | 457S     | NaN  | 363200      |
| 354 | LITTLE YORK RD & SHADY LN (21016141) | 414S     | NaN  | 363378      |
| 486 | 7345 OAK HILL DR                     | 535S     | NaN  | 363511      |
| 487 | 7345 OAK HILL DR                     | 535S     | NaN  | 363512      |
| 488 | 4983 DUMFRIES DR                     | 531U     | NaN  | 363513      |
| 489 | 7353 OAK HILL DR                     | 535S     | NaN  | 363515      |
| 490 | 5107 GRAPE ST                        | 531Q     | NaN  | 363516      |
| 492 | 6839 RICHWOOD RD                     | 534V     | NaN  | 363518      |
| 493 | 5103 PEASE ST, HOUSTON, 77023        | 494X     | NaN  | 363519      |
| 495 | 4934 HULL ST                         | 534N     | NaN  | 363521      |
| 499 | 7730 BOBBITT LN, HOUSTON, 77055      | 451X     | NaN  | 363525      |
| 500 | 918 LOPER AVE, HOUSTON, 77017        | 536J     | NaN  | 363527      |
| 501 | 1620 NEVADA ST                       | 492R     | NaN  | 363528      |
| 502 | 401 STRATFORD ST                     | 493S     | NaN  | 363529      |
| 503 | 2529 SPENCE ST, HOUSTON, 77093       | 454N     | NaN  | 363530      |
| 508 | 511 E 14TH ST, HOUSTON, 77008        | 453W     | NaN  | 363535      |
| 509 | 1638 LATEXO DR                       | 452E     | NaN  | 363536      |
| 510 | 1102 CALIFORNIA ST, HOUSTON, 77006   | 493S     | NaN  | 363537      |
| 511 | 1103 MISSOURI ST, HOUSTON, 77006     | 493S     | NaN  | 363538      |
| 514 | 700 WESTMONT DR                      | 496H     | NaN  | 363541      |
| 515 | 3000 TURNER DR, HOUSTON, 77093       | 454A     | NaN  | 363542      |
| 518 | 3803 W LAKE HOUSTON PKWY             | 297X     | NaN  | 363545      |
| 521 | 3514 ASKEW ST                        | 534L     | NaN  | 363548      |

|     | address                                      | map_page | shop | workorderid |
|-----|----------------------------------------------|----------|------|-------------|
| 522 | 3514 ASKEW ST                                | 534L     | NaN  | 363549      |
| 524 | 6221 MAIN ST                                 | 533A     | NaN  | 363551      |
| 526 | 422 TRUMAN ST, HOUSTON, 77018                | 452M     | NaN  | 363553      |
| 527 | 3602 W PINE BROOK WAY                        | 578Y     | NaN  | 363554      |
| 528 | 15607 WINTER BRIAR DR                        | 571X     | NaN  | 363555      |
| 529 | 6301 BOBBY BURNS ST, HOUSTON, 77028          | 455P     | NaN  | 363556      |
| 530 | 607 E 7TH 1/2 ST, HOUSTON, 77007             | 493A     | NaN  | 363557      |
| 531 | 8101 N BAYOU DR (GLENBROOK PARK)             | 535Q     | NaN  | 363558      |
| 532 | 7820 SCHNEIDER ST, HOUSTON, 77093            | 453M     | NaN  | 363559      |
| 533 | 8237 CREEKBEND DR                            | 530X     | NaN  | 363560      |
| 535 | 7514 IMOGENE ST, HOUSTON, 77074              | 530Q     | NaN  | 363562      |
| 536 | 7 ADAMS ST                                   | 494T     | NaN  | 363563      |
| 537 | 7743 WILEYVALE RD                            | 454L     | NaN  | 363564      |
| 538 | 1615 NEVADA ST                               | 492R     | NaN  | 363565      |
| 548 | 7514 IMOGENE ST                              | 530Q     | NaN  | 363575      |
| 549 | 7514 IMOGENE ST                              | 530Q     | NaN  | 363576      |
| 551 | 5642 TRAFALGAR DR                            | 571Q     | NaN  | 363579      |
| 553 | 5823 QUEENSLOCH DR                           | 531S     | NaN  | 363581      |
| 554 | 3400 REED RD                                 | 573B     | NaN  | 363582      |
| 555 | 5214 NASSAU RD                               | 534K     | NaN  | 363583      |
| 556 | 4118 BEGGS ST, HOUSTON, 77009                | 453Y     | NaN  | 363584      |
| 789 | 3517 REED RD (JOHNSON NEIGHBOORHOOD LIBRARY) | 533P     | NaN  | 363837      |
| 796 | 6019 ENGLAND ST                              | 533G     | NaN  | 363845      |
| 824 | 4700 CLINTON DR (JAPHET CREEK PARK)          | 494L     | NaN  | 363874      |

|      | address                                           | map_page | shop | workorderid |
|------|---------------------------------------------------|----------|------|-------------|
| 1419 | 301 KNIPP RD                                      | 490K     | NaN  | 364486      |
| 2125 | 2026 RANKIN RD (R-000265-0117-3)                  | 373G     | NaN  | 365206      |
| 2394 | 12223 MELROSE PARK RD (MELROSE PARK)              | 413S     | NaN  | 365477      |
| 2700 | KINGWOOD DR (FROM ROYAL FOREST TO LOOP 494)       | 336A     | NaN  | 365796      |
| 2702 | KINGWOOD DR (FROM ROYAL FOREST TO CHESTNUT RIDGE) | 336A     | NaN  | 365798      |
| 2703 | KINGWOOD DR (FROM ROYAL FOREST AND CHESTNUT RI... | 336A     | NaN  | 365799      |
| 2704 | KINGWOOD DR (FROM CHESTNUTRIDGE TO GREEN OAK)     | 336A     | NaN  | 365800      |
| 2705 | KINGWOOD DR (FROM CHESTNUT RIDGE TO GREEN OAK)    | 336A     | NaN  | 365801      |
| 2706 | KINGWOOD DR (FROM GREEN OAK TO LADBROOK)          | 336A     | NaN  | 365802      |
| 2711 | 1859 KINGWOOD DR, KINGWOOD, TEXAS, 77339          | 336A     | NaN  | 365807      |
| 2713 | KINGWOOD DR (EAST OF TRAILWOOD VILLAGE)           | 336A     | NaN  | 365809      |
| 2714 | KINGWOOD DR (EAST OF TRAILWOOD VILLAGE)           | 336A     | NaN  | 365810      |

In [149... *#Displaying address and map\_page columns to help determine shop value. Reviewing last 22 records.*

```
df_columns = df[['address', 'map_page', 'shop', 'workorderid']]
df_columns[df_columns['shop'].isnull()].tail(22)
```

Out[149...

|      | address                                           | map_page | shop | workorderid |
|------|---------------------------------------------------|----------|------|-------------|
| 2717 | KINGWOOD DR (FROM FOREST SHORES TO KING FOREST... | 336A     | NaN  | 365813      |
| 2719 | KINGWOOD DR (FROM W LAKE CRESCENT TO E LAKE CR... | 336A     | NaN  | 365815      |
| 2720 | KINGWOOD DR (WEST OF WOOS ESTATES)                | 336A     | NaN  | 365816      |
| 2725 | KINGWOOD DR (FROM WOODS ESTATE TO VALLEY MANOR    | 336A     | NaN  | 365821      |
| 2731 | KINGWOOD DR (FROM VALLEY MANOR TO KINGWOOD HS)    | 336A     | NaN  | 365828      |
| 2733 | 4360 KINGWOOD DR                                  | 336A     | NaN  | 365830      |
| 2735 | KINGWOOD DR (FROM KINGWOOD HIGH SCHOOL TO W LA... | 336A     | NaN  | 365832      |
| 2745 | NORTHPARK DR (FROM W LAKE HOUSTON PKWY TO LAKE... | 297T     | NaN  | 365842      |
| 2746 | 3310 NORTHPARK DR                                 | 297T     | NaN  | 365843      |
| 2747 | NORTHPARK DR (FROM LAKE PINE TO DEERBROOK)        | 297T     | NaN  | 365844      |
| 2748 | NORTHPARK DR (FROM DEERBROOK TO BROOKDALE)        | 297T     | NaN  | 365845      |
| 2760 | NORTHPARK DR (FROM BROOKDALE TO WOODBRIDGE)       | 297T     | NaN  | 365857      |
| 2767 | NORTHPARK DR (FROM WOODBRIDGE TO BASSINGHAM)      | 297T     | NaN  | 365864      |
| 2768 | NORTHPARK DR (FROM BASSINGHAM TO MEADOW SPRINGS)  | 297T     | NaN  | 365865      |
| 2769 | NORTHPARK DR (FROM FOREST BLUFF TO MEADOW SPRI... | 297T     | NaN  | 365866      |
| 2770 | NORTHPARK DR (FROM FOREST BLUFF TO WOODLAND HI... | 297T     | NaN  | 365867      |
| 2775 | NORTHPARK DR (FROM GLADE VALLEY TO HIDDEN PINES)  | 297T     | NaN  | 365872      |
| 2776 | NORTHPARK DR (FROM GLADE VALLEY TO WOODLAND HI... | 297T     | NaN  | 365873      |
| 2777 | NORTHPARK DR (FROM HIDDEN PINES TO CITY LIMIT ... | 297T     | NaN  | 365874      |
| 2782 | 1310 NORTHPARK DR KINGWOOD TX                     | 297T     | NaN  | 365879      |
| 2783 | NORTHPARK DR (FROM 494 TO RUSSELL PALMER)         | 297T     | NaN  | 365880      |
| 2784 | NORTHPARK DR (FROM 494 TO RUSSELL PALMER)         | 297T     | NaN  | 365881      |

```
In [150... #Replacing null values in shop with known values
df.iloc[144, 9] = 'E'
```

```
df.iloc[212, 9] = 'A'
df.iloc[354, 9] = 'E'
df.iloc[486, 9] = 'E'
df.iloc[487, 9] = 'E'
df.iloc[488, 9] = 'E'
df.iloc[489, 9] = 'E'
df.iloc[490, 9] = 'E'
df.iloc[492, 9] = 'E'
df.iloc[493, 9] = 'E'
df.iloc[495, 9] = 'E'
df.iloc[499, 9] = 'E'
df.iloc[500, 9] = 'E'
df.iloc[501, 9] = 'E'
df.iloc[502, 9] = 'E'
df.iloc[503, 9] = 'E'
df.iloc[508, 9] = 'E'
df.iloc[509, 9] = 'E'
df.iloc[510, 9] = 'E'
df.iloc[511, 9] = 'E'
df.iloc[514, 9] = 'E'
df.iloc[515, 9] = 'E'
df.iloc[518, 9] = 'C'
df.iloc[521, 9] = 'E'
df.iloc[522, 9] = 'I'
df.iloc[524, 9] = 'G'
df.iloc[526, 9] = 'B'
df.iloc[527, 9] = 'E'
df.iloc[528, 9] = 'B'
df.iloc[529, 9] = 'A'
df.iloc[530, 9] = 'A'
df.iloc[531, 9] = 'I'
df.iloc[532, 9] = 'E'
df.iloc[533, 9] = 'A'
df.iloc[535, 9] = 'A'
df.iloc[536, 9] = 'A'
df.iloc[537, 9] = 'E'
df.iloc[538, 9] = 'D'
df.iloc[548, 9] = 'D'
df.iloc[549, 9] = 'D'
df.iloc[551, 9] = 'J'
df.iloc[553, 9] = 'A'
df.iloc[554, 9] = 'E'
```

```
df.iloc[555, 9] = 'A'  
df.iloc[556, 9] = 'E'  
df.iloc[789, 9] = 'A'  
df.iloc[796, 9] = 'E'  
df.iloc[824, 9] = 'H'  
df.iloc[1419, 9] = 'G'  
df.iloc[2125, 9] = 'B'  
df.iloc[2394, 9] = 'I'  
df.iloc[2700, 9] = 'E'  
df.iloc[2702, 9] = 'E'  
df.iloc[2703, 9] = 'E'  
df.iloc[2704, 9] = 'E'  
df.iloc[2705, 9] = 'E'  
df.iloc[2706, 9] = 'E'  
df.iloc[2711, 9] = 'E'  
df.iloc[2713, 9] = 'E'  
df.iloc[2714, 9] = 'E'  
df.iloc[2717, 9] = 'E'  
df.iloc[2719, 9] = 'E'  
df.iloc[2720, 9] = 'E'  
df.iloc[2725, 9] = 'E'  
df.iloc[2731, 9] = 'E'  
df.iloc[2733, 9] = 'E'  
df.iloc[2735, 9] = 'E'  
df.iloc[2745, 9] = 'E'  
df.iloc[2746, 9] = 'E'  
df.iloc[2747, 9] = 'E'  
df.iloc[2748, 9] = 'E'  
df.iloc[2760, 9] = 'E'  
df.iloc[2767, 9] = 'E'  
df.iloc[2768, 9] = 'E'  
df.iloc[2769, 9] = 'E'  
df.iloc[2770, 9] = 'E'  
df.iloc[2775, 9] = 'E'  
df.iloc[2776, 9] = 'E'  
df.iloc[2777, 9] = 'E'  
df.iloc[2782, 9] = 'E'  
df.iloc[2783, 9] = 'E'  
df.iloc[2784, 9] = 'E'
```

In [151... *#Confirming null values in shop have been replaced with the known values*

```
df[df['shop'].isnull()]
```

```
Out[151...  workorderid  description  priority  status  submit_to  proj_start_date  proj_finish_date  actual_start  actual_finish  shop  ...  date_initi
```

0 rows × 24 columns

```
In [152... #Calculating the sum of all missing values for each column to confirm that all relevent missing values h
df.isna().sum()
```

```
Out[152... workorderid          0
description          0
priority            0
status             0
submit_to          0
proj_start_date     0
proj_finish_date    0
actual_start        0
actual_finish        2
shop               0
map_page           0
address            0
location           0
category           0
date_initiated      0
project            1562
total_cost          0
comments           137
zone               1
request_ids         0
resolution          0
sr_problem_code     0
sr_description      0
sr_priority         0
dtype: int64
```

```
In [153... #Displaying a dataframe of null values in the actual_finish column
df[df['actual_finish'].isnull()]
```

Out[153...

|      | workorderid | description                                     | priority | status | submit_to         | proj_start_date        | proj_finish_date       | actual_start           | actual_finish | shop |
|------|-------------|-------------------------------------------------|----------|--------|-------------------|------------------------|------------------------|------------------------|---------------|------|
| 892  | 363943      | REMOVAL &<br>STUMP<br>GRINDING BY<br>CONTRACTOR | 3        | ISSUED | UF,<br>CONTRACTOR | 2022-05-05<br>04:57 PM | 2022-06-04<br>04:57 PM | 2022-08-05<br>12:00 PM | NaN           | H    |
| 1429 | 364496      | REMOVAL &<br>STUMP<br>GRINDING BY<br>CONTRACTOR | 3        | ISSUED | UF,<br>CONTRACTOR | 2022-06-14<br>02:26 PM | 2022-07-14<br>02:26 PM | 2026-01-09<br>12:00 PM | NaN           | H    |

2 rows × 24 columns

In [154... *#Replacing null values in actual\_finish with values from proj\_finish\_date*  
`df['actual_finish'] = df['actual_finish'].fillna(df['proj_finish_date'])`

In [155... *#Displaying a dataframe of null values in the zone column*  
`df[df['zone'].isnull()]`

Out[155...

|     | workorderid | description                       | priority | status | submit_to        | proj_start_date        | proj_finish_date       | actual_start           | actual_finish          | shop | ... | d |
|-----|-------------|-----------------------------------|----------|--------|------------------|------------------------|------------------------|------------------------|------------------------|------|-----|---|
| 797 | 363846      | EMERGENCY<br>TRIM BY IN-<br>HOUSE | 1        | CLOSED | UF, IN-<br>HOUSE | 2022-04-27<br>02:45 PM | 2022-04-28<br>02:45 PM | 2022-04-27<br>02:45 PM | 2022-04-27<br>10:42 PM | C    | ... |   |

1 rows × 24 columns

In [156... *#Replacing null values in zone with known values*  
`df['zone'] = df['zone'].fillna('4, ZONE')`

In [157... *#Calculating the sum of all missing values for each column to confirm that all relevent missing values h*  
`df.isna().sum()`

```
Out[157... workorderid      0
           description     0
           priority        0
           status          0
           submit_to       0
           proj_start_date  0
           proj_finish_date 0
           actual_start    0
           actual_finish   0
           shop            0
           map_page        0
           address         0
           location        0
           category        0
           date_initiated  0
           project         1562
           total_cost      0
           comments        137
           zone            0
           request_ids     0
           resolution      0
           sr_problem_code  0
           sr_description   0
           sr_priority      0
           dtype: int64
```

```
In [158... #Reviewing data types for all columns. Checking for date fields that need to be converted to date data t
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2826 entries, 0 to 2825
Data columns (total 24 columns):
#   Column                Non-Null Count  Dtype
---  -
0   workorderid           2826 non-null   int64
1   description            2826 non-null   object
2   priority               2826 non-null   int64
3   status                 2826 non-null   object
4   submit_to              2826 non-null   object
5   proj_start_date        2826 non-null   object
6   proj_finish_date       2826 non-null   object
7   actual_start           2826 non-null   object
8   actual_finish          2826 non-null   object
9   shop                   2826 non-null   object
10  map_page                2826 non-null   object
11  address                 2826 non-null   object
12  location                2826 non-null   object
13  category                2826 non-null   object
14  date_initiated          2826 non-null   object
15  project                 1264 non-null   object
16  total_cost              2826 non-null   float64
17  comments                2689 non-null   object
18  zone                    2826 non-null   object
19  request_ids             2826 non-null   object
20  resolution              2826 non-null   object
21  sr_problem_code         2826 non-null   object
22  sr_description          2826 non-null   object
23  sr_priority             2826 non-null   float64
dtypes: float64(2), int64(2), object(20)
memory usage: 530.0+ KB

```

```

In [159... #Splitting 'proj_start_date' field into two fields to separate date and time
df[['projected_start_date', 'projected_start_time']] = df['proj_start_date'].str.split(' ', expand=True, n
df

```

Out[159...

|      | workorderid | description                         | priority | status | submit_to         | proj_start_date        | proj_finish_date       | actual_start           | actual_finish          | shc |
|------|-------------|-------------------------------------|----------|--------|-------------------|------------------------|------------------------|------------------------|------------------------|-----|
| 0    | 362924      | TRIM BY<br>CONTRACTOR               | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>07:24 AM | 2022-02-03<br>07:24 AM | 2022-04-01<br>12:00 PM | 2022-05-11<br>12:00 PM |     |
| 1    | 362925      | EMERGENCY<br>DEBRIS PICK-UP         | 1        | CLOSED | UF, IN-HOUSE      | 2022-01-04<br>07:45 AM | 2022-01-05<br>07:45 AM | 2022-01-04<br>07:45 AM | 2022-01-11<br>10:39 AM |     |
| 2    | 362928      | REMOVAL BY<br>CONTRACTOR            | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>08:00 AM | 2022-02-03<br>08:00 AM | 2022-04-01<br>12:00 PM | 2023-01-31<br>12:00 PM |     |
| 3    | 362932      | DEBRIS PICK-UP                      | 2        | CLOSED | UF, IN-HOUSE      | 2022-01-04<br>08:54 AM | 2022-01-14<br>08:54 AM | 2022-01-04<br>08:54 AM | 2022-01-20<br>10:27 AM |     |
| 4    | 362933      | EMERGENCY<br>REMOVAL BY<br>IN-HOUSE | 1        | CLOSED | UF, IN-HOUSE      | 2022-01-04<br>08:55 AM | 2022-01-05<br>08:55 AM | 2022-01-04<br>08:55 AM | 2022-01-07<br>02:05 PM |     |
| ...  | ...         | ...                                 | ...      | ...    | ...               | ...                    | ...                    | ...                    | ...                    | ... |
| 2821 | 365918      | TREE/SEEDLING<br>PLANTING           | 4        | CLOSED | UF,<br>CONTRACTOR | 2022-12-29<br>07:14 AM | 2023-12-29<br>07:14 AM | 2022-12-29<br>07:14 AM | 2022-12-29<br>12:00 PM |     |
| 2822 | 365919      | TRIM BY<br>CONTRACTOR               | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-12-29<br>07:31 AM | 2023-01-28<br>07:31 AM | 2023-03-13<br>12:00 PM | 2025-03-21<br>09:08 AM |     |

|      | workorderid | description               | priority | status | submit_to         | proj_start_date        | proj_finish_date       | actual_start           | actual_finish          | shc |
|------|-------------|---------------------------|----------|--------|-------------------|------------------------|------------------------|------------------------|------------------------|-----|
| 2823 | 365920      | TREE/SEEDLING<br>PLANTING | 4        | CLOSED | UF,<br>CONTRACTOR | 2022-12-30<br>08:15 AM | 2023-12-30<br>08:15 AM | 2022-12-30<br>08:15 AM | 2022-12-30<br>12:00 PM |     |
| 2824 | 365921      | TRIM BY<br>CONTRACTOR     | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-12-30<br>12:45 PM | 2023-01-29<br>12:45 PM | 2023-03-13<br>12:00 PM | 2023-04-11<br>12:00 PM |     |
| 2825 | 365922      | TRIM BY IN-<br>HOUSE      | 2        | CLOSED | UF, IN-HOUSE      | 2022-12-30<br>01:13 PM | 2023-01-09<br>01:13 PM | 2022-12-30<br>01:13 PM | 2023-01-19<br>02:25 PM |     |

2826 rows × 26 columns

```
In [160... #Splitting 'proj_finish_date' field into two fields to separate date and time
df[['projected_finish_date', 'projected_finish_time']] = df['proj_finish_date'].str.split(' ', expand=True)
df
```

Out[160...

|             | workorderid | description                         | priority | status | submit_to         | proj_start_date        | proj_finish_date       | actual_start           | actual_finish          | shc |
|-------------|-------------|-------------------------------------|----------|--------|-------------------|------------------------|------------------------|------------------------|------------------------|-----|
| <b>0</b>    | 362924      | TRIM BY<br>CONTRACTOR               | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>07:24 AM | 2022-02-03<br>07:24 AM | 2022-04-01<br>12:00 PM | 2022-05-11<br>12:00 PM |     |
| <b>1</b>    | 362925      | EMERGENCY<br>DEBRIS PICK-UP         | 1        | CLOSED | UF, IN-HOUSE      | 2022-01-04<br>07:45 AM | 2022-01-05<br>07:45 AM | 2022-01-04<br>07:45 AM | 2022-01-11<br>10:39 AM |     |
| <b>2</b>    | 362928      | REMOVAL BY<br>CONTRACTOR            | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>08:00 AM | 2022-02-03<br>08:00 AM | 2022-04-01<br>12:00 PM | 2023-01-31<br>12:00 PM |     |
| <b>3</b>    | 362932      | DEBRIS PICK-UP                      | 2        | CLOSED | UF, IN-HOUSE      | 2022-01-04<br>08:54 AM | 2022-01-14<br>08:54 AM | 2022-01-04<br>08:54 AM | 2022-01-20<br>10:27 AM |     |
| <b>4</b>    | 362933      | EMERGENCY<br>REMOVAL BY<br>IN-HOUSE | 1        | CLOSED | UF, IN-HOUSE      | 2022-01-04<br>08:55 AM | 2022-01-05<br>08:55 AM | 2022-01-04<br>08:55 AM | 2022-01-07<br>02:05 PM |     |
| ...         | ...         | ...                                 | ...      | ...    | ...               | ...                    | ...                    | ...                    | ...                    | ... |
| <b>2821</b> | 365918      | TREE/SEEDLING<br>PLANTING           | 4        | CLOSED | UF,<br>CONTRACTOR | 2022-12-29<br>07:14 AM | 2023-12-29<br>07:14 AM | 2022-12-29<br>07:14 AM | 2022-12-29<br>12:00 PM |     |
| <b>2822</b> | 365919      | TRIM BY<br>CONTRACTOR               | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-12-29<br>07:31 AM | 2023-01-28<br>07:31 AM | 2023-03-13<br>12:00 PM | 2025-03-21<br>09:08 AM |     |
| <b>2823</b> | 365920      | TREE/SEEDLING<br>PLANTING           | 4        | CLOSED | UF,<br>CONTRACTOR | 2022-12-30<br>08:15 AM | 2023-12-30<br>08:15 AM | 2022-12-30<br>08:15 AM | 2022-12-30<br>12:00 PM |     |
| <b>2824</b> | 365921      | TRIM BY<br>CONTRACTOR               | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-12-30<br>12:45 PM | 2023-01-29<br>12:45 PM | 2023-03-13<br>12:00 PM | 2023-04-11<br>12:00 PM |     |
| <b>2825</b> | 365922      | TRIM BY IN-<br>HOUSE                | 2        | CLOSED | UF, IN-HOUSE      | 2022-12-30<br>01:13 PM | 2023-01-09<br>01:13 PM | 2022-12-30<br>01:13 PM | 2023-01-19<br>02:25 PM | C   |

| workorderid | description | priority | status | submit_to | proj_start_date | proj_finish_date | actual_start | actual_finish | shc |
|-------------|-------------|----------|--------|-----------|-----------------|------------------|--------------|---------------|-----|
|-------------|-------------|----------|--------|-----------|-----------------|------------------|--------------|---------------|-----|

2826 rows × 28 columns

```
In [161... #Splitting 'actual_start' field into two fields to separate date and time
df[['actual_start_date', 'actual_start_time']] = df['actual_start'].str.split(' ', expand=True, n=1)
df
```

Out[161...

|             | workorderid | description                         | priority | status | submit_to         | proj_start_date        | proj_finish_date       | actual_start           | actual_finish          | shc |
|-------------|-------------|-------------------------------------|----------|--------|-------------------|------------------------|------------------------|------------------------|------------------------|-----|
| <b>0</b>    | 362924      | TRIM BY<br>CONTRACTOR               | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>07:24 AM | 2022-02-03<br>07:24 AM | 2022-04-01<br>12:00 PM | 2022-05-11<br>12:00 PM |     |
| <b>1</b>    | 362925      | EMERGENCY<br>DEBRIS PICK-UP         | 1        | CLOSED | UF, IN-HOUSE      | 2022-01-04<br>07:45 AM | 2022-01-05<br>07:45 AM | 2022-01-04<br>07:45 AM | 2022-01-11<br>10:39 AM |     |
| <b>2</b>    | 362928      | REMOVAL BY<br>CONTRACTOR            | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>08:00 AM | 2022-02-03<br>08:00 AM | 2022-04-01<br>12:00 PM | 2023-01-31<br>12:00 PM |     |
| <b>3</b>    | 362932      | DEBRIS PICK-UP                      | 2        | CLOSED | UF, IN-HOUSE      | 2022-01-04<br>08:54 AM | 2022-01-14<br>08:54 AM | 2022-01-04<br>08:54 AM | 2022-01-20<br>10:27 AM |     |
| <b>4</b>    | 362933      | EMERGENCY<br>REMOVAL BY<br>IN-HOUSE | 1        | CLOSED | UF, IN-HOUSE      | 2022-01-04<br>08:55 AM | 2022-01-05<br>08:55 AM | 2022-01-04<br>08:55 AM | 2022-01-07<br>02:05 PM |     |
| ...         | ...         | ...                                 | ...      | ...    | ...               | ...                    | ...                    | ...                    | ...                    | ... |
| <b>2821</b> | 365918      | TREE/SEEDLING<br>PLANTING           | 4        | CLOSED | UF,<br>CONTRACTOR | 2022-12-29<br>07:14 AM | 2023-12-29<br>07:14 AM | 2022-12-29<br>07:14 AM | 2022-12-29<br>12:00 PM |     |
| <b>2822</b> | 365919      | TRIM BY<br>CONTRACTOR               | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-12-29<br>07:31 AM | 2023-01-28<br>07:31 AM | 2023-03-13<br>12:00 PM | 2025-03-21<br>09:08 AM |     |
| <b>2823</b> | 365920      | TREE/SEEDLING<br>PLANTING           | 4        | CLOSED | UF,<br>CONTRACTOR | 2022-12-30<br>08:15 AM | 2023-12-30<br>08:15 AM | 2022-12-30<br>08:15 AM | 2022-12-30<br>12:00 PM |     |
| <b>2824</b> | 365921      | TRIM BY<br>CONTRACTOR               | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-12-30<br>12:45 PM | 2023-01-29<br>12:45 PM | 2023-03-13<br>12:00 PM | 2023-04-11<br>12:00 PM |     |
| <b>2825</b> | 365922      | TRIM BY IN-<br>HOUSE                | 2        | CLOSED | UF, IN-HOUSE      | 2022-12-30<br>01:13 PM | 2023-01-09<br>01:13 PM | 2022-12-30<br>01:13 PM | 2023-01-19<br>02:25 PM | C   |

| workorderid | description | priority | status | submit_to | proj_start_date | proj_finish_date | actual_start | actual_finish | shc |
|-------------|-------------|----------|--------|-----------|-----------------|------------------|--------------|---------------|-----|
|-------------|-------------|----------|--------|-----------|-----------------|------------------|--------------|---------------|-----|

2826 rows × 30 columns

```
In [162... #Splitting 'actual_finish' field into two fields to separate date and time
df[['actual_finish_date', 'actual_finish_time']] = df['actual_finish'].str.split(' ', expand=True, n=1)
df
```

Out[162...

|             | workorderid | description                   | priority | status | submit_to      | proj_start_date     | proj_finish_date    | actual_start        | actual_finish       | shc |
|-------------|-------------|-------------------------------|----------|--------|----------------|---------------------|---------------------|---------------------|---------------------|-----|
| <b>0</b>    | 362924      | TRIM BY CONTRACTOR            | 3        | CLOSED | UF, CONTRACTOR | 2022-01-04 07:24 AM | 2022-02-03 07:24 AM | 2022-04-01 12:00 PM | 2022-05-11 12:00 PM |     |
| <b>1</b>    | 362925      | EMERGENCY DEBRIS PICK-UP      | 1        | CLOSED | UF, IN-HOUSE   | 2022-01-04 07:45 AM | 2022-01-05 07:45 AM | 2022-01-04 07:45 AM | 2022-01-11 10:39 AM |     |
| <b>2</b>    | 362928      | REMOVAL BY CONTRACTOR         | 3        | CLOSED | UF, CONTRACTOR | 2022-01-04 08:00 AM | 2022-02-03 08:00 AM | 2022-04-01 12:00 PM | 2023-01-31 12:00 PM |     |
| <b>3</b>    | 362932      | DEBRIS PICK-UP                | 2        | CLOSED | UF, IN-HOUSE   | 2022-01-04 08:54 AM | 2022-01-14 08:54 AM | 2022-01-04 08:54 AM | 2022-01-20 10:27 AM |     |
| <b>4</b>    | 362933      | EMERGENCY REMOVAL BY IN-HOUSE | 1        | CLOSED | UF, IN-HOUSE   | 2022-01-04 08:55 AM | 2022-01-05 08:55 AM | 2022-01-04 08:55 AM | 2022-01-07 02:05 PM |     |
| ...         | ...         | ...                           | ...      | ...    | ...            | ...                 | ...                 | ...                 | ...                 | ... |
| <b>2821</b> | 365918      | TREE/SEEDLING PLANTING        | 4        | CLOSED | UF, CONTRACTOR | 2022-12-29 07:14 AM | 2023-12-29 07:14 AM | 2022-12-29 07:14 AM | 2022-12-29 12:00 PM |     |
| <b>2822</b> | 365919      | TRIM BY CONTRACTOR            | 3        | CLOSED | UF, CONTRACTOR | 2022-12-29 07:31 AM | 2023-01-28 07:31 AM | 2023-03-13 12:00 PM | 2025-03-21 09:08 AM |     |
| <b>2823</b> | 365920      | TREE/SEEDLING PLANTING        | 4        | CLOSED | UF, CONTRACTOR | 2022-12-30 08:15 AM | 2023-12-30 08:15 AM | 2022-12-30 08:15 AM | 2022-12-30 12:00 PM |     |
| <b>2824</b> | 365921      | TRIM BY CONTRACTOR            | 3        | CLOSED | UF, CONTRACTOR | 2022-12-30 12:45 PM | 2023-01-29 12:45 PM | 2023-03-13 12:00 PM | 2023-04-11 12:00 PM |     |
| <b>2825</b> | 365922      | TRIM BY IN-HOUSE              | 2        | CLOSED | UF, IN-HOUSE   | 2022-12-30 01:13 PM | 2023-01-09 01:13 PM | 2022-12-30 01:13 PM | 2023-01-19 02:25 PM |     |

2826 rows × 32 columns

In [163...

```
#Splitting 'date_initiated' field into two fields to separate date and time
df[['initiated_date', 'initiated_time']] = df['date_initiated'].str.split(' ', expand=True, n=1)
df
```

Out[163...

|      | workorderid | description                   | priority | status | submit_to      | proj_start_date     | proj_finish_date    | actual_start        | actual_finish       | shc |
|------|-------------|-------------------------------|----------|--------|----------------|---------------------|---------------------|---------------------|---------------------|-----|
| 0    | 362924      | TRIM BY CONTRACTOR            | 3        | CLOSED | UF, CONTRACTOR | 2022-01-04 07:24 AM | 2022-02-03 07:24 AM | 2022-04-01 12:00 PM | 2022-05-11 12:00 PM |     |
| 1    | 362925      | EMERGENCY DEBRIS PICK-UP      | 1        | CLOSED | UF, IN-HOUSE   | 2022-01-04 07:45 AM | 2022-01-05 07:45 AM | 2022-01-04 07:45 AM | 2022-01-11 10:39 AM |     |
| 2    | 362928      | REMOVAL BY CONTRACTOR         | 3        | CLOSED | UF, CONTRACTOR | 2022-01-04 08:00 AM | 2022-02-03 08:00 AM | 2022-04-01 12:00 PM | 2023-01-31 12:00 PM |     |
| 3    | 362932      | DEBRIS PICK-UP                | 2        | CLOSED | UF, IN-HOUSE   | 2022-01-04 08:54 AM | 2022-01-14 08:54 AM | 2022-01-04 08:54 AM | 2022-01-20 10:27 AM |     |
| 4    | 362933      | EMERGENCY REMOVAL BY IN-HOUSE | 1        | CLOSED | UF, IN-HOUSE   | 2022-01-04 08:55 AM | 2022-01-05 08:55 AM | 2022-01-04 08:55 AM | 2022-01-07 02:05 PM |     |
| ...  | ...         | ...                           | ...      | ...    | ...            | ...                 | ...                 | ...                 | ...                 | ... |
| 2821 | 365918      | TREE/SEEDLING PLANTING        | 4        | CLOSED | UF, CONTRACTOR | 2022-12-29 07:14 AM | 2023-12-29 07:14 AM | 2022-12-29 07:14 AM | 2022-12-29 12:00 PM |     |
| 2822 | 365919      | TRIM BY CONTRACTOR            | 3        | CLOSED | UF, CONTRACTOR | 2022-12-29 07:31 AM | 2023-01-28 07:31 AM | 2023-03-13 12:00 PM | 2025-03-21 09:08 AM |     |
| 2823 | 365920      | TREE/SEEDLING PLANTING        | 4        | CLOSED | UF, CONTRACTOR | 2022-12-30 08:15 AM | 2023-12-30 08:15 AM | 2022-12-30 08:15 AM | 2022-12-30 12:00 PM |     |
| 2824 | 365921      | TRIM BY CONTRACTOR            | 3        | CLOSED | UF, CONTRACTOR | 2022-12-30 12:45 PM | 2023-01-29 12:45 PM | 2023-03-13 12:00 PM | 2023-04-11 12:00 PM |     |
| 2825 | 365922      | TRIM BY IN-HOUSE              | 2        | CLOSED | UF, IN-HOUSE   | 2022-12-30 01:13 PM | 2023-01-09 01:13 PM | 2022-12-30 01:13 PM | 2023-01-19 02:25 PM |     |

2826 rows × 34 columns

In [164...

```

#Converting all date fields to datetime data type
df['proj_start_date'] = pd.to_datetime(df['proj_start_date'], dayfirst=False)
df['proj_finish_date'] = pd.to_datetime(df['proj_finish_date'], dayfirst=False)
df['projected_start_date'] = pd.to_datetime(df['projected_start_date'], dayfirst=False)
df['projected_finish_date'] = pd.to_datetime(df['projected_finish_date'], dayfirst=False)
df['actual_start'] = pd.to_datetime(df['actual_start'], dayfirst=False)
df['actual_start_date'] = pd.to_datetime(df['actual_start_date'], dayfirst=False)
df['actual_finish_date'] = pd.to_datetime(df['actual_finish_date'], dayfirst=False)

```

```
df['actual_finish'] = pd.to_datetime(df['actual_finish'], dayfirst=False)
df['initiated_date'] = pd.to_datetime(df['initiated_date'], dayfirst=False)
df
```

Out[164]...

|      | workorderid | description                         | priority | status | submit_to         | proj_start_date        | proj_finish_date       | actual_start           | actual_finish          | shc |
|------|-------------|-------------------------------------|----------|--------|-------------------|------------------------|------------------------|------------------------|------------------------|-----|
| 0    | 362924      | TRIM BY<br>CONTRACTOR               | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>07:24:00 | 2022-02-03<br>07:24:00 | 2022-04-01<br>12:00:00 | 2022-05-11<br>12:00:00 |     |
| 1    | 362925      | EMERGENCY<br>DEBRIS PICK-UP         | 1        | CLOSED | UF, IN-HOUSE      | 2022-01-04<br>07:45:00 | 2022-01-05<br>07:45:00 | 2022-01-04<br>07:45:00 | 2022-01-11<br>10:39:00 |     |
| 2    | 362928      | REMOVAL BY<br>CONTRACTOR            | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-01-04<br>08:00:00 | 2022-02-03<br>08:00:00 | 2022-04-01<br>12:00:00 | 2023-01-31<br>12:00:00 |     |
| 3    | 362932      | DEBRIS PICK-UP                      | 2        | CLOSED | UF, IN-HOUSE      | 2022-01-04<br>08:54:00 | 2022-01-14<br>08:54:00 | 2022-01-04<br>08:54:00 | 2022-01-20<br>10:27:00 |     |
| 4    | 362933      | EMERGENCY<br>REMOVAL BY<br>IN-HOUSE | 1        | CLOSED | UF, IN-HOUSE      | 2022-01-04<br>08:55:00 | 2022-01-05<br>08:55:00 | 2022-01-04<br>08:55:00 | 2022-01-07<br>14:05:00 |     |
| ...  | ...         | ...                                 | ...      | ...    | ...               | ...                    | ...                    | ...                    | ...                    | ... |
| 2821 | 365918      | TREE/SEEDLING<br>PLANTING           | 4        | CLOSED | UF,<br>CONTRACTOR | 2022-12-29<br>07:14:00 | 2023-12-29<br>07:14:00 | 2022-12-29<br>07:14:00 | 2022-12-29<br>12:00:00 |     |
| 2822 | 365919      | TRIM BY<br>CONTRACTOR               | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-12-29<br>07:31:00 | 2023-01-28<br>07:31:00 | 2023-03-13<br>12:00:00 | 2025-03-21<br>09:08:00 |     |
| 2823 | 365920      | TREE/SEEDLING<br>PLANTING           | 4        | CLOSED | UF,<br>CONTRACTOR | 2022-12-30<br>08:15:00 | 2023-12-30<br>08:15:00 | 2022-12-30<br>08:15:00 | 2022-12-30<br>12:00:00 |     |
| 2824 | 365921      | TRIM BY<br>CONTRACTOR               | 3        | CLOSED | UF,<br>CONTRACTOR | 2022-12-30<br>12:45:00 | 2023-01-29<br>12:45:00 | 2023-03-13<br>12:00:00 | 2023-04-11<br>12:00:00 |     |
| 2825 | 365922      | TRIM BY IN-<br>HOUSE                | 2        | CLOSED | UF, IN-HOUSE      | 2022-12-30<br>13:13:00 | 2023-01-09<br>13:13:00 | 2022-12-30<br>13:13:00 | 2023-01-19<br>14:25:00 |     |

2826 rows × 34 columns

In [165]...

```
#Verifying data types for all fields
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2826 entries, 0 to 2825
Data columns (total 34 columns):
#   Column                Non-Null Count  Dtype
---  -
0   workorderid            2826 non-null   int64
1   description            2826 non-null   object
2   priority               2826 non-null   int64
3   status                2826 non-null   object
4   submit_to             2826 non-null   object
5   proj_start_date       2826 non-null   datetime64[ns]
6   proj_finish_date      2826 non-null   datetime64[ns]
7   actual_start          2826 non-null   datetime64[ns]
8   actual_finish         2826 non-null   datetime64[ns]
9   shop                  2826 non-null   object
10  map_page              2826 non-null   object
11  address               2826 non-null   object
12  location              2826 non-null   object
13  category              2826 non-null   object
14  date_initiated        2826 non-null   object
15  project               1264 non-null   object
16  total_cost            2826 non-null   float64
17  comments              2689 non-null   object
18  zone                  2826 non-null   object
19  request_ids           2826 non-null   object
20  resolution            2826 non-null   object
21  sr_problem_code       2826 non-null   object
22  sr_description        2826 non-null   object
23  sr_priority           2826 non-null   float64
24  projected_start_date  2826 non-null   datetime64[ns]
25  projected_start_time  2826 non-null   object
26  projected_finish_date 2826 non-null   datetime64[ns]
27  projected_finish_time 2826 non-null   object
28  actual_start_date     2826 non-null   datetime64[ns]
29  actual_start_time     2826 non-null   object
30  actual_finish_date    2826 non-null   datetime64[ns]
31  actual_finish_time    2826 non-null   object
32  initiated_date        2826 non-null   datetime64[ns]
33  initiated_time        2826 non-null   object
dtypes: datetime64[ns](9), float64(2), int64(2), object(21)
memory usage: 750.8+ KB

```

```
In [166... #Saving cleaned data set as new csv file  
df.to_csv('My_Projects/Urban_Forestry/Cityworks_WO/Cityworks_WOs_2022_cleaned.csv', index=False)
```

```
In [ ]:
```